

From Pinski & Narin to the JIRC algorithm

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1 Introduction

Pinski-Narin (PN,1976) presented an approach to journal influence that accounted for the difference level of influence of journals when they provide citations to the journal of interest. The approach leads to an iteration over all journals to find the influence of each journal, following the evolution of a Markov chain of cross-citations. The work of Pinski-Narin was developed further to construct the Article Influence Score (Clarivate).

The Reputable Citation (RC) (Safon et al., 2025) algorithm is an extension of this class of algorithms through using (adding) prestige weights to each journal and citing institution.

2 The citation matrix

Following PN, let's define the "unit" as a journal j which is a member of a set of selected journals J , where $j = 1 \dots M$.

Let $C = (c_{ij})$ be a matrix whose elements are the sum of references emitted by the articles published in journal i (rows) that become citations received by any of the articles published in journal j (columns) during the target publication window. If we ignore journal self-citations, a common practice, then $c_{ii} = 0$.

Define the sum of all references from i as

$$s_i = \sum_{j=1}^M c_{ij} \quad (1)$$

The matrix C can be normalised as the row stochastic matrix P where

$$P = (p_{ij}) = (c_{ij}/s_i) \quad (2)$$

in which case citations from a journal are normalized by the total number of citations coming from that journal to the other journals in the ensemble.

A stochastic matrix is a transition probability Markov matrix. The repeated product P^n represents a step-wise chain from an initial to a final state after n steps. If the matrix P is irreducible, there is a single asymptotic final state; if not, there is no unique final state. It is desirable for computational reasons that the stochastic matrix P is irreducible. We will assume that in what follows.

3 Influence Factor (Pinski & Narin - PN)

Define the journal influence weight vector $w = (w_j)$ as a column vector of influence weights for each journal in the set. The influence weight of a journal averages the weighted number of citations from every journal to that journal so that, ultimately, journals with “above average” influence weight are those that receive more citations from journals with high influence weight and vice versa. This leads to an implicit equation for w

$$w_j = \sum_{i=1}^N p_{ij} w_i \quad (3)$$

$$w = P^T w \quad (4)$$

This equation defines the influence weight for a journal. The influence weight per publication w' is

$$w'_j = \frac{(P^T w)_j}{a_j} \quad (5)$$

where a_j is the number of articles published in journal j in the time period selected.

PN uses the iteration (called the “Power method”)

$$w(k) = \frac{P^T w^{(k-1)}}{|P^T w^{(k-1)}|} \quad (6)$$

that converges to the principal eigenvalue and the desired influence vector.

4 The Journal-Institution Reputation Circulation algorithm (JIRC)

The dataset now contains the ensemble of journals, $j = 1 \dots M$, as well as a collection of institutions, $i = 1 \dots N$ with a sizable number of papers in the field and period under analysis.

When computing journal influence scores, JIRC will inject values of citing article prestige, v_i , based on institutions assessments derived from the dataset. Conversely, to computing institutional influence scores, JIRC will inject values of citing article prestige, w_j , based on journal assessments derived from the same dataset.

We will formulate the problem as a (re)circulation of reputation. The idea is to go from institutions to institutions through journals and vice versa.

The problem can be stated as follows.

Let's introduce citation matrix A , from institutions ($i = 1 \dots N$) to journals ($j = 1 \dots M$), $A = (a_{ij})$, whose elements are the sum of references emitted by the articles published by institution i (rows) that become citations received by

any of the articles published in journal j (columns) during the target publication window.

Let's again define the sum of all references from institution i as

$$s_i = \sum_{j=1}^M c_{ij} \quad (7)$$

The matrix A can be normalised as the row stochastic matrix P_A , where

$$P_A = (p_{ij}) = (c_{ij}/s_i) \quad (8)$$

in which case citations from an institution are normalized by the total number of citations coming from that institution to all the journals in the dataset.

We also introduce citation matrix B , from journals to institutions, $B = (c_{ji})$, whose elements are the sum of references emitted by the articles published by journal j (rows) that become citations received by any of the articles published in institution i (columns) during the target publication window.

$$s_j = \sum_{i=1}^N c_{ji} \quad (9)$$

The matrix B can be normalised as the row stochastic matrix P_B , where

$$P_B = (p_{ji}) = (c_{ji}/s_j) \quad (10)$$

in which case citations from a journal are normalized by the total number of citations coming from that journal to all the institutions in the dataset.

The influence vectors are connected by the two matrices. P_A^T is $M \times N$, whereas P_B^T is $N \times M$. Both influence column vectors, w , $M \times 1$, for journals and v , $N \times 1$, for institutions, receive reputation shots from each other, $w = P_A^T v$, $v = P_B^T w$.

Hence, we can now treat the reputation transfer full circle, as

$$v = P_B^T P_A^T v \quad (11)$$

$$w = P_A^T P_B^T w \quad (12)$$

Matrices $P_A^T P_B^T$ and $P_B^T P_A^T$ are both square, although they have, in general, different sizes: $M \times M$ and $N \times N$, respectively.

However, it is well-known that they share all the nonzero eigenvalues, the largest in particular, implying that once we find the eigenvector v corresponding to the largest eigenvalue of $P_B^T P_A^T$, we know that the eigenvector corresponding to the largest eigenvalue of $P_A^T P_B^T$ will be $w = P_A^T v$, and vice versa.

The equation $v = P_B^T P_A^T v$ defines the influence weight for an institution. The influence weight per publication v' is

$$v'_i = \frac{(P_B^T P_A^T v)_i}{a_i} \quad (13)$$

where a_i is the number of articles published by institution in the field and time period selected.

JIRC will also use the Power method either for institutions or journals, since both will produce the same result:

$$v(k) = \frac{P_B^T P_A^T v^{(k-1)}}{|P_B^T P_A^T v^{(k-1)}|} \quad (14)$$

that converges to the principal eigenvalue and the desired influence vector. The equation $w = P_A^T v$ will render the eigenvector for the journal to journal transfer of reputation,

$$w'_j = \frac{(P_A^T v)_j}{a_j} \quad (15)$$

References

Pinski-Narin (1976) Citation influence for journal aggregates of scientific publications: Theory, with application to the literature of physics. Information Processing and Management.

Safon-Docampo-Cram (2025) Screening articles by citation reputation. Quantitative Science Studies 6: 405–420.